

Experimental phase equilibria study and thermodynamic modelling of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ ternary system in equilibrium with metallic copper

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ABSTRACT

Phase equilibria studies were undertaken on the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper using equilibration and quenching technique followed by the electron probe X-ray microanalysis (EPMA). The liquidus of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system, including the two 2-liquid miscibility gaps in the system as well as the cuprite (Cu₂O), corundum (Al₂O₃), delafossite (CuAlO₂), tridymite and cristobalite (Si_{1-x}(CuAl)_xO₂), and mullite (Al_{6+2x}Si_{2-2x}O_{13-x}) primary phase fields has been characterized. New experimental results together with phase equilibria and thermodynamic literature data were used to obtain a self-consistent set of parameters of the thermodynamic model for all phases. The liquidus projection of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system was presented for the first time in the full range of temperatures and compositions. It was found that the simultaneous dissolution of a small amount of alumina and copper oxide (CuAlO₂) in SiO₂ phases stabilize cristobalite over tridymite at low temperatures.

1. Introduction

The “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper is an important part of copper smelting, converting and recycling slags in use by industry today [1]. Silica (SiO₂) can be found in copper concentrates and is the main flux in slags used in the pyrometallurgical production of copper [2]. Alumina (Al₂O₃) is usually introduced into the process with ore concentrate or recycled waste from electrical and electronic equipment (WEEE); its addition was found to lower copper solubilities in fayalite slag in equilibrium with matte [3]. Besides pyrometallurgy, the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system forms the basis for the manufacturing of the low-expansion aluminosilicate glasses [4] and describes the interaction of copper oxide-containing melts with alumina-silica refractories [5]. The aim of the present study is to provide a clearer understanding of the chemical properties of melts taking part in these processes through the characterisation of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper.

The present study is an important part of the integrated experimental and thermodynamic modelling research program on the Cu-Pb-Zn-Fe-Si-Ca-Al-Mg-O system [6], representing the principal components of non-ferrous pyrometallurgical primary production and recycling slags [7]. The new experimental data on this important subsystem will enable this and the larger multicomponent systems to be described using

chemical thermodynamic models. Phase equilibria data and other thermodynamic properties of these systems will then provide an important foundation for the improved understanding and, importantly, prediction of the behaviour of these slags in the industrial processes.

The oxidation state of Cu can continuously change from 0 to 1⁺ and 2⁺ above 1400 °C in the Cu-O system [8]. Presence of metallic phase makes the contribution of Cu²⁺ negligible, while AlO_{1.5} and SiO₂ suppress the formation of Cu⁰ in slag. The Cu⁺ as a predominant species was confirmed by previous slag-metal distribution studies as a function of oxygen partial pressure (pO₂) [9–11]. Therefore, quotation marks for “CuO_{0.5}” in slag are used in the present study to indicate insignificant deviation from Cu⁺ towards either Cu⁰ or Cu²⁺.

No binary compounds were reported in the experimental studies of the “CuO_{0.5}”-SiO₂ system in equilibrium with metallic copper [12–17]. The “CuO_{0.5}”-AlO_{1.5} system in equilibrium with metallic copper and the AlO_{1.5}-SiO₂ system, in addition to the endmembers cuprite (Cu₂O), corundum (Al₂O₃) and tridymite/cristobalite (SiO₂), contains delafossite (CuAlO₂) [12,18,19] and mullite (Al_{6+2x}Si_{2-2x}O_{13-x}) [20–30] phases, respectively.

Two experimental studies of the liquidus surface of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper were found in the literature [14,19]. Hidayat et al. [14] measured liquidus isotherms at 1200 and 1300 °C in the tridymite (SiO₂) primary phase field. Chen

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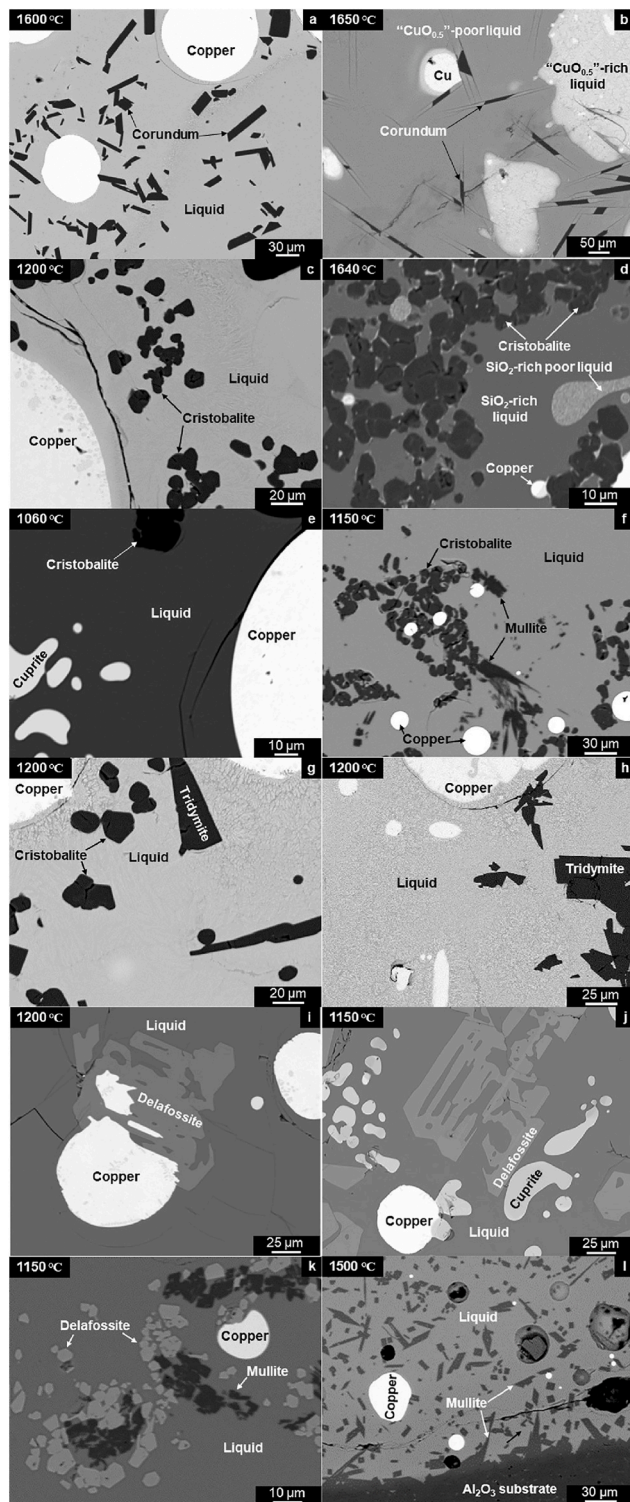


Fig. 1. Backscattered SEM micrographs of the quenched “CuO_{0.5}”-AlO_{1.5}-SiO₂ samples in equilibrium with metallic copper: (a) liquid-corundum (Al₂O₃)-metal; (b) 2 liquids-corundum (Al₂O₃)-metal; (c) liquid-cristobalite (Si_{1-x}(CuAl)_xO₂)-metal; (d) 2 liquids-cristobalite (SiO₂)-metal (e) liquid-cristobalite (Si_{1-x}(CuAl)_xO₂)-cuprite (Cu₂O)-metal; (f) liquid-cristobalite (Si_{1-x}(CuAl)_xO₂)-mullite (Al_{6+2x}Si_{2-2x}O_{13-x})-metal; (g) liquid-cristobalite (Si_{1-x}(CuAl)_xO₂)-tridymite (Si_{1-x}(CuAl)_xO₂)-metal; (h) liquid-tridymite (Si_{1-x}(CuAl)_xO₂)-metal; (i) liquid-delafofossite (CuAlO₂)-metal; (j) liquid-delafofossite (CuAlO₂)-cuprite (Cu₂O)-metal; (k) liquid-delafofossite (CuAlO₂)-mullite (Al_{6+2x}Si_{2-2x}O_{13-x})-metal; (l) liquid-mullite (Al_{6+2x}Si_{2-2x}O_{13-x})-metal.

and Zhao [19] reported equilibria in the primary phase fields of corundum (Al₂O₃), cuprite (Cu₂O), delafossite (CuAlO₂) and tridymite (SiO₂) at temperatures between 1150 and 1300 °C; their results agree with the findings by Hidayat et al. [14]. The current study provides experimental description of the liquidus surface of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper over the whole range of compositions at multiple temperatures.

2. Experimental technique

The experimental procedure, using equilibration and quenching followed by EPMA, and apparatus used in the present study were similar to those previously reported by the authors [31–34]. The initial mixtures were made from high-purity powders Cu₂O, Al₂O₃ and SiO₂ (>99.9 wt% purity, supplied by Alfa Aesar, MA, USA or Heysham, Great Britain). To ensure that metallic copper was present at the final equilibrium in all samples, 10–20 wt% excess Cu powder (99.9 wt% purity; supplied by Alfa Aesar, MA, USA) was added to the oxide mixtures. Less than 0.3 g of mixture was used in each equilibration experiment. The initial compositions of the mixtures were selected so that one or more crystalline phases were present in equilibrium with liquid slag. The initial predictions of the final phase composition were made for every sample using Factsage [35] and internal thermodynamic database [6]. The volume fraction of solids was targeted to be below 50%, and preferably about 10%, to achieve rapid equilibration and satisfactory quenching of the liquid slag to amorphous material. It was found that the solid phases acted as heterogeneous nucleation centers, and a minimum distance of 10–20 μm between solid grains was necessary to ensure that an amorphous slag of uniform composition, unaffected by dendritic crystal growth during quenching, was obtained. The correlation between the proportion of the solid phases present in equilibrium and the distance between solid grains was described previously by the authors [36]. An iterative procedure involving preliminary experiments was often required to achieve the targeted phase proportions for a given equilibration temperature and composition, since the exact liquidus compositions at a given temperature were not initially known. Only homogenous slag areas, that is, those not affected by the growth of the “quench crystals”, were analysed and included in the present study.

The mixtures were pressed into pellets using a tool steel die before being mounted on a substrate. Silica ampoules (5 cm long, 1.3 cm outer diameter, 99.999%, supplied by GY, Jiangsu, China) and alumina crucibles (1.5 cm long, 1 cm outer diameter, 99.7 wt% purity, supplied by Xiamen Wintrustek Advance Materials Co., LTD, China) were used as sample support substrates for equilibration, depending on the target composition. Pt foil was found to be unsuitable to be used to study the phase equilibria in the Cu-containing systems due to the possible formation of the Cu-Pt solution [37] which would lead to the destruction of the substrate material and contamination of the system with Pt. At the same time the ceramic crucibles were found to be reliable to be used as substrates [14,36,38]. The precipitation of the primary phase on the substrate prevents the excessive dissolution of the substrate material allowing the bulk composition of the system to stay constant during the equilibration time [39,40]. The ceramic crucibles should then be chosen so not to contaminate the studied system. Silica ampoules were used for the experiments targeting the primary phase field of tridymite/cristobalite (Si_{1-x}(CuAl)_xO₂), while alumina crucibles for primary phase field of corundum (Al₂O₃). Either of the substrates were suitable for the experiments in the primary phase fields of delafossite (CuAlO₂) and mullite (Al_{6+2x}Si_{2-2x}O_{13-x}). In these cases it was found that in the first minutes of the equilibration the substrate was covered with the precipitated primary phase becoming isolated from the liquid slag which prevented the dissolution of the substrate material in the liquid phase (see Fig. 11). Similar behaviour was observed for the experiments in the primary phase field of tridymite/cristobalite (Si_{1-x}(CuAl)_xO₂); the intermediate layer of tridymite/cristobalite (Si_{1-x}(CuAl)_xO₂) was formed on the border between the SiO₂ ampoule and the liquid slag. The composition

of this layer was similar to that of tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) solids, however it was not identified whether it has precipitated from the solution or formed by the interaction between the substrate and the liquid.

Each equilibration experiment was carried out in an electrical-resistance heated vertical tube furnace with an impervious recrystallized alumina (30-mm inner diameter) reaction tube with silicon carbide (SiC) heating element; the upper temperature limit for the furnace operation was 1700 °C. The samples were placed immediately adjacent to a working thermocouple (type B) that was encased in a recrystallized alumina sheath in the uniform hot zone of the furnace to monitor the actual sample temperature. The working thermocouple was calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia). The overall absolute temperature accuracy of the experiment was estimated to be ± 3 K.

The samples were suspended in the hot zone of the furnace by Kanthal support wire (Fe-Cr-Al alloy, 0.7 (Kanthal D) or 1-mm (Kanthal A-1) diameter; melting temperature 1500 °C; suitable for operating up to $p\text{O}_2 = 1$ atm; supplied by Vulcan Stainless, VIC, Australia). For temperatures higher than 1500 °C, a 20–25 cm Pt or Pt-Rh wire (99.99 wt%, 0.5 mm diameter, supplied by Johnson Matthey Ltd, VIC, Australia) was used to extend the support. For selected experiments, the samples were preheated for 15–60 min at 10–200 °C above the equilibration temperature to form a homogeneous liquid slag or pre-cooled to promote the formation of multiple solid phases rather than supercooled slag. This was followed by the equilibration at the final target temperature for the required time. At the end of the equilibration process, the samples were released and rapidly quenched into the CaCl_2 brine solution (−20 °C). Brine solution was chosen over water due to the higher cooling rates [41, 42]. The samples were then washed thoroughly in water and ethanol, dried on a hot plate, mounted in epoxy resin and cross-sectioned and polished using conventional metallographic techniques.

The cross-sectioned samples were examined by optical microscopy, carbon-coated, and the compositions of the phases were measured by EPMA with the wavelength-dispersive detectors (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan) operated with 15 kV accelerating voltage and 20 nA probe current using the standard Duncumb–Philibert atomic number, absorption, and fluorescence (ZAF) correction supplied by JEOL [43–45]. Copper (Cu) (for Cu and “ $\text{CuO}_{0.5}$ ” measurements), corundum (Al_2O_3) (for $\text{AlO}_{1.5}$ measurements), hematite (Fe_2O_3) (for $\text{FeO}_{1.5}$ measurements, to check possible impurity) and wollastonite (CaSiO_3) (for SiO_2 measurements) (supplied by Charles M. Taylor Co., Stanford, CA) were used as the EPMA standards. The measurement uncertainty corresponded to the use of metallic copper as a standard for oxide measurements did not exceed 0.2%. The concentrations of metal cations were measured with EPMA; no data on the oxidation states of the metal cations in slag were obtained. For presentation purposes, the copper oxide concentrations were then re-calculated as “ $\text{CuO}_{0.5}$ ”; this oxidation state was selected according to the information on the slags in equilibrium with metal in the Cu-Si-O system [13]. It was shown [46] that the increase from fully focused electron beam (zero probe diameter) to 50 μm decreased the uncertainty of the copper slags compositions measured with EPMA and did not affect the accuracy of measurements. The probe diameter was selected depending on the size of the phase under investigation. The slag compositions reported in this study were measured mainly using 50 μm probe diameter. The compositions of the phases reported in the present study are the average values of a number (from 10 to 20, usually 15) of independent measurements from different regions of the samples rather than single measurements.

The accuracy of the compositional measurements was improved on that obtained using the standard JEOL ZAF correction following an approach similar to that described previously by the authors [17, 47–49]. This involved using stoichiometric compounds of known composition as secondary internal standards. The $\text{AlO}_{1.5}$ concentration in the delafossite (CuAlO_2 ; coexisting in equilibrium with “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 oxide

liquid) measured with EPMA was found to be overestimated by 2.8 mol.% over its stoichiometric value (50 mol.% $\text{AlO}_{1.5}$) after the standard JEOL ZAF correction. Previous study [17] also showed the overestimation of SiO_2 measured with EPMA in the stoichiometric dehydrated diopside (CuSiO_3 ; single crystals of natural mineral from Altyn-Tyube diopside deposit, Altyn-Tyube area, Karaganda, Kazakhstan [17]) by 1.8 mol.%. The accuracy of the measurement then was improved by the following corrections:

$$x_{\text{CuO}_x}^{\text{corrected}} = x_{\text{CuO}_x} + 0.072 \times x_{\text{SiO}_2} x_{\text{CuO}_x} + 0.116 \times x_{\text{AlO}_{1.5}} x_{\text{CuO}_x}; \quad (1)$$

$$x_{\text{AlO}_{1.5}}^{\text{corrected}} = x_{\text{AlO}_{1.5}} - 0.116 \times x_{\text{AlO}_{1.5}} x_{\text{CuO}_x}; \quad (2)$$

$$x_{\text{SiO}_2}^{\text{corrected}} = x_{\text{SiO}_2} - 0.072 \times x_{\text{SiO}_2} x_{\text{CuO}_x}, \quad (3)$$

where $x(\text{CuO}_x, \text{AlO}_{1.5}, \text{SiO}_2)$ are the cation molar fractions after the standard JEOL ZAF corrections, and $x(\text{CuO}_x, \text{AlO}_{1.5}, \text{SiO}_2)^{\text{corrected}}$ are the corrected cation molar fractions respectively.

The overestimation of concentrations of some elements such as Fe, Cu, Zn in tridymite or cristobalite (SiO_2) within the Fe, Cu, Zn containing matrix (slag) measured with EPMA, due to the secondary fluorescence by the X-rays originated from the continuum background spectra of the inelastic scattering of electrons (bremsstrahlung), was previously reported [50,51]. The same effect was observed by the authors in previous studies (Fe [52,53], Cu [17,36,39], Zn [39,53,54]). The concentrations of the Fe, Cu and Zn oxides initially measured to be solid solution in tridymite and cristobalite (SiO_2) in the slag containing samples, were proven to be overestimated values by undertaking EPMA measurements of these elements in unreacted pellets consisted of the 10–100 μm pure SiO_2 particles mechanically pressed within the Fe_2O_3 , Cu or ZnO powder matrix. This bremsstrahlung fluorescence effect was only observed for the measurements using characteristic X-ray lines with the energies above 5 keV; the X-rays from the continuum background with these energies have high (>100 μm) penetration ability through the SiO_2 or other particles of comparable density into the surrounding matrix (e.g. slag) thus exciting the characteristic lines of Fe, Cu, Zn and therefore generating the incorrect apparent overestimated values of concentrations of these elements in such particles [50,51]. Measurements using the characteristic X-ray lines with low energy do not have this effect due to the strong absorption of that spectrum range of the background X-rays [50,51].

Preliminary measurements of the Cu oxide concentrations in phases with known compositions using L_{α} -line (0.93 keV) indicated significant uncertainty in the corresponding standard JEOL ZAF corrections [36]. The custom additional correction for the concentrations measured with the K_{α} -lines (equations (1)–(3)) had already been developed and tested by the authors [17,47–49]. The following procedure therefore was developed and used for measurements of the Cu concentration in the low-Cu-oxide phases. The copper oxide concentrations were measured in the same areas of several well-quenched slag samples using both Cu K_{α} and L_{α} lines and values obtained were found to be different from each other with the $(\text{Cu } K_{\alpha})/(\text{Cu } L_{\alpha})$ ratio being >1 for the whole measured range of concentrations. The following correction was developed and used to re-calculate the concentrations of copper oxide measured in tridymite/cristobalite (SiO_2) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) with L_{α} lines into correct “K” concentrations [36]:

$$x_{\text{CuO}_{0.5},\text{K}} = -0.0013 (x_{\text{CuO}_{0.5},\text{L}})^2 + 1.1261 x_{\text{CuO}_{0.5},\text{L}} \quad (4)$$

where $x_{\text{CuO}_{0.5},\text{L}}$ are the Cu oxide concentration values measured with EPMA using L_{α} -values after the standard JEOL ZAF correction in mol.%, and $x_{\text{CuO}_{0.5},\text{K}}$ are the apparent “pseudo” K_{α} Cu oxide concentration values derived from $x_{\text{CuO}_{0.5},\text{L}}$ also in mol.%. equations (1)–(3) were then applied using the $x_{\text{CuO}_{0.5},\text{K}}$ values derived from $x_{\text{CuO}_{0.5},\text{L}}$.

For the purposes of the tridymite vs. cristobalite (SiO_2) identification, the samples were designed to contain >90% solids and therefore were not suitable for the accurate EPMA measurements of the liquid

Table 1Experimental measurements of phase compositions for the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper.

Mixture #	Substrate	Preheat/precool	Final equilibration	Time	Phases present	Compositions (mol.%)		
		T, °C		h		“CuO0.5”	AlO1.5	SiO2
Initial mixtures								
1	–					0.71	6.0	93.3
2	–					1.7	9.5	88.8
3	–					8.0	17.5	74.5
4	–					11.9	2.7	85.4
5	–					12.5	19.5	68.0
6	–					15.0	45.0	40.0
7	–					19.5	39.8	40.7
8	–					20.2	36.1	43.7
9	–					22.3	3.1	74.5
10	–					23.3	12.0	64.7
11	–					24.5	5.5	70.0
12	–					25.7	27.0	47.3
13	–					26.8	1.4	71.8
14	–					27.3	35.4	37.3
15	–					27.8	2.2	70.0
16	–					28.6	43.0	28.4
17	–					29.5	2.7	67.9
18	–					29.7	6.7	63.6
19	–					30.0	42.0	28.0
20	–					30.8	1.3	67.8
21	–					31.0	48.4	20.5
22	–					31.5	3.5	65.0
23	–					34.0	15.5	50.5
24	–					34.1	0.96	65.0
25	–					38.0	35.0	27.0
26	–					40.5	12.5	47.0
27	–					46.0	4.0	50.0
28	–					47.4	5.5	47.0
29	–					50.9	23.8	25.3
30	–					52.0	33.0	15.0
31	–					61.5	7.0	31.5
32	–					64.5	5.8	29.7
33	–					65.0	24.0	11.0
34	–					69.0	4.4	26.7
35	–					73.3	3.2	23.5
36	–					77.5	2.2	20.3
37	–					82.5	0.89	16.6
Corundum (Al2O3) liquidus								
21	Al2O3	–	1700	0.25	Liquid	28.0 ± 1.0	45.7 ± 0.6	26.3 ± 0.5
					Corundum	0.04L	0.10	99.9
16	Al2O3	–	1600	1	Liquid	26.6 ± 0.3	39.1 ± 0.3	34.3 ± 0.2
					Corundum	0.50F	99.50	0.00
30	Al2O3	–	1600	0.75	Liquid	46.1 ± 0.3	35.6 ± 0.2	18.2 ± 0.1
					Corundum	0.87F	99.10	0.03
30	Al2O3		1600	0.75	Liquid	49.7 ± 0.2	33.5 ± 0.1	16.8 ± 0.1
					Corundum	0.88F	99.09	0.04
14	Al2O3	–	1500	1	Liquid	25.0 ± 0.1	33.1 ± 0.1	42.0 ± 0.2
					Corundum	0.05L	99.82	0.13
19	Al2O3	1520	1500	2	Liquid	31.0 ± 0.4	33.6 ± 0.2	35.3 ± 0.4
					Corundum	1.2F	98.7	0.06
25	Al2O3	1530	1500	1	Liquid	36.8 ± 0.2	33.5 ± 0.2	29.7 ± 0.1
					Corundum	1.3F	98.7	0.02
30	Al2O3	1520	1500	1	Liquid	54.5 ± 0.3	26.7 ± 0.2	18.8 ± 0.1
					Corundum	1.4F	98.6	0.01
30	Al2O3	1520	1500	1	Liquid	56.5 ± 0.1	25.5 ± 0.2	18.0 ± 0.1
					Corundum	1.2F	98.84	0.00
33	Al2O3	1500	1450	1	Liquid	69.5 ± 0.4	17.2 ± 0.3	13.3 ± 0.2
					Corundum	1.5F	98.5	0.01
25	Al2O3	1450	1400	2	Liquid	42.1 ± 0.2	27.6 ± 0.1	30.3 ± 0.1
					Corundum	0.62F	99.38	0.00
25	Al2O3	1450	1400	2	Liquid	43.8 ± 0.1	27.0 ± 0.1	29.3 ± 0.2
					Corundum	0.64F	99.36	0.00
30	Al2O3	1450	1400	2	Liquid	53.5 ± 0.7	24.5 ± 0.4	22.0 ± 0.3
					Corundum	1.4F	98.6	0.05
19	Al2O3	1450	1400	2	Liquid	56.8 ± 0.8	22.7 ± 0.5	20.4 ± 0.3
					Corundum	1.4F	98.6	0.05
2 Liquids + Corundum (Al2O3)								
33	Al2O3	–	1650	1	AlO1.5-rich liquid	49.5 ± 0.2	36.7 ± 0.1	13.8 ± 0.1
					AlO1.5-poor liquid	95.7 ± 0.3	3.2 ± 0.3	1.2 ± 0.1
					Corundum	0.65F	99.32	0.02
33	Al2O3	–	1550	0.75	AlO1.5-rich liquid	61.5 ± 1.3	24.8 ± 0.6	13.7 ± 0.7
					AlO1.5-poor liquid	90.7 ± 1.2	5.2 ± 1.1	4.1 ± 0.4

(continued on next page)

Table 1 (continued)

Mixture #	Substrate	Preheat/precool	Final equilibration	Time	Phases present	Compositions (mol.%)		
		T, °C		h		“CuO0.5”	AlO1.5	SiO2
Corundum (Al2O3) + Mullite (Al6+2xSi2-2xO13-x)					Corundum	1.4F	98.5	0.03
7	Al2O3	–	1700	0.25	Liquid Corundum Mullite	18.2 ± 0.1 0.07L 0.63L	43.5 ± 1.1 0.15 22.1	38.3 ± 1.0 99.8 77.3
Cristobalite (Si1-x(CuAl)xO2) liquidus					Liquid	12.3 ± 0.1	4.6 ± 0.1	83.1 ± 0.1
9	SiO2	–	1600	0.5	Cristobalite	–	–	–
9	SiO2	–	1600	0.5	Liquid Cristobalite	16.0 ± 0.5 –	4.4 ± 0.0 –	79.6 ± 0.5 –
15	SiO2	1610	1600	1.5	Liquid Cristobalite	38.8 ± 0.5 2.0F	3.4 ± 0.0 1.1	57.8 ± 0.5 96.9
15	SiO2	1610	1600	1.5	Liquid Cristobalite	41.2 ± 0.7 2.0F	3.4 ± 0.0 1.1	55.4 ± 0.6 96.9
1	SiO2	1600	1580	24	Liquid Cristobalite	1.1 ± 0.1 0.06L	7.7 ± 0.6 0.07 99	91.2 ± 0.6 99.98
22	SiO2	1570	1550	2	Liquid Cristobalite	42.6 ± 0.4 –	4.5 ± 0.0 –	53.0 ± 0.3 –
22	SiO2	1570	1550	2	Liquid Cristobalite	44.4 ± 0.5 –	4.3 ± 0.1 –	51.3 ± 0.5 –
22	SiO2	1570	1550	2	Liquid Cristobalite	46.8 ± 0.4 1.6F	4.1 ± 0.0 0.5	49.1 ± 0.4 97.9
11	SiO2	1520	1500	3	Liquid Cristobalite	13.5 ± 0.4 1.4L	8.9 ± 0.1 1.5	77.6 ± 0.4 97.1
18	SiO2	1520	1500	1	Liquid Cristobalite	15.6 ± 0.4 –	8.8 ± 0.2 –	75.7 ± 0.5 –
11	SiO2	1520	1500	2	Liquid Cristobalite	20.0 ± 1.6 1.0L	8.3 ± 0.4 1.1	71.7 ± 1.4 97.9
4	SiO2	1520	1500X	4	Liquid Cristobalite	21.4 ± 1.5 1.1L	7.4 ± 0.2 1.0	71.2 ± 1.3 97.1
18	SiO2	1520	1500	2	Liquid Cristobalite	25.0 ± 1.4 0.91L	7.6 ± 0.1 0.82	67.4 ± 1.3 98.3
18	SiO2	1520	1500	4	Liquid Cristobalite	26.6 ± 0.9 –	7.2 ± 0.1 –	66.2 ± 0.8 –
27	SiO2	1520	1500	2	Liquid Cristobalite	47.6 ± 0.6 1.8F	4.8 ± 0.1 0.57	47.5 ± 0.5 97.6
18	SiO2	1420	1400	1	Liquid Cristobalite	16.6 ± 0.5 –	12.4 ± 0.2 –	71.0 ± 0.4 –
18	SiO2	1420	1400	1	Liquid Cristobalite	20.2 ± 0.4 –	11.7 ± 0.1 –	68.1 ± 0.4 –
4	SiO2	1420	1400X	4	Liquid Cristobalite	30.8 ± 0.3 1.1L	9.5 ± 0.1 1.3	59.6 ± 0.4 98.3
11	SiO2	1420	1400	2	Liquid Cristobalite	32.9 ± 0.8 1.36L	9.6 ± 0.2 1.2	57.5 ± 0.6 97.4
18	SiO2	1420	1400	2	Liquid Cristobalite	33.5 ± 0.5 1.4L	9.5 ± 0.1 1.5	57.0 ± 0.4 97.0
18	SiO2	1420	1400	4	Liquid Cristobalite	36.7 ± 0.4 0.79L	9.1 ± 0.1 0.97	54.2 ± 0.4 98.2
28	SiO2	1420	1400	3	Liquid Cristobalite	44.0 ± 0.3 –	7.9 ± 0.1 –	48.2 ± 0.2 –
28	SiO2	1420	1400	3	Liquid Cristobalite	47.6 ± 0.2 0.78L	7.2 ± 0.1 0.75	45.1 ± 0.2 98.5
11	SiO2	1320	1300	2	Liquid Cristobalite	28.4 ± 0.7 1.8L	13.1 ± 0.0 2.0	58.5 ± 0.7 96.2
10	SiO2	1220	1200	3	Liquid Cristobalite	20.3 ± 0.3 –	17.6 ± 0.2 –	62.1 ± 0.4 –
32	SiO2	1250	1200	2	Liquid Cristobalite	62.0 ± 0.2 0.82L	6.6 ± 0.1 0.99	31.4 ± 0.1 98.2
32	SiO2	–	1200	4	Liquid Cristobalite	62.9 ± 0.4 0.80L	6.6 ± 0.1 0.79	30.5 ± 0.3 98.4
17	SiO2	1220	1200X	4	Liquid Cristobalite	64.8 ± 0.2 0.89L	5.5 ± 0.1 0.86	29.7 ± 0.2 98.3
34	SiO2	–	1200	4	Liquid Cristobalite	67.0 ± 0.3 0.85L	5.6 ± 0.1 0.83	27.4 ± 0.2 98.3
23	SiO2	1120/1080	1100	168	Liquid Cristobalite	22.9 ± 0.1 –	19.3 ± 0.1 –	57.8 ± 0.1 –
23	SiO2	1120/1080	1100	168	Liquid Cristobalite	23.9 ± 0.1 0.7F	19.0 ± 0.1 0.83	57.2 ± 0.1 98.4
2 Liquids + Cristobalite (Si1-x(CuAl)xO2)					SiO2-rich liquid SiO2-poor liquid Cristobalite	13.3 ± 0.1 53.2 ± 1.0 0.31L	1.7 ± 0.0 1.5 ± 0.1 0.30	85.1 ± 0.1 45.3 ± 1.0 99.39
13	SiO2	–	1660	1	SiO2-rich liquid	16.9 ± 0.2	2.1 ± 0.0	81.0 ± 0.2

(continued on next page)

Table 1 (continued)

Mixture #	Substrate	Preheat/precool	Final equilibration	Time h	Phases present	Compositions (mol.%)		
		T, °C				“CuO0.5”	AlO1.5	SiO2
13	SiO2	–	1640	1	SiO2-poor liquid	51.5 ± 0.9	1.8 ± 0.1	46.7 ± 0.9
					Cristobalite	0.75F	0.40	98.9
					SiO2-rich liquid	21.7 ± 1.2	2.3 ± 0.1	75.8 ± 1.2
					SiO2-poor liquid	46.7 ± 0.6	2.5 ± 0.1	51.0 ± 0.6
26	SiO2	1140/1080	1120	168	Cristobalite	0.48L	0.47	99.1
					Cuprite (Cu2O)			
					Liquid	34.0 ± 0.3	15.8 ± 0.1	50.2 ± 0.3
					Cristobalite	2.4F	1.4	96.2
23	SiO2	1120/1050	1060	168	Cuprite	99.99	0.01	0.00
					Liquid	26.7 ± 0.1	19.0 ± 0.1	54.3 ± 0.1
					Cristobalite	1.8L	2.2	96.1
					Cuprite	100.00	0.00	0.00
2	SiO2	1530	1500	24	Cristobalite (Si1-x(CuAl)xO2) + Mullite (Al6+2xSi2-2xO13-x)			
					Liquid	4.3 ± 0.1	12.3 ± 0.2	83.3 ± 0.3
					Cristobalite	0.12L	0.12L	99.76
					Mullite	0.08L	73.5	26.5
3	SiO2	1450	1400	18	Liquid	10.5 ± 0.1	16.0 ± 0.1	73.5 ± 0.1
					Cristobalite	1.1L	1.2	97.7
					Mullite	0.40L	76.0	23.6
					Liquid	17.7 ± 0.2	19.9 ± 0.1	62.5 ± 0.2
5	SiO2	1250	1150	72	Cristobalite	2.7L	2.9	94.4
					Mullite	0.64L	75.8	23.5
					Liquid	19.2 ± 0.2	20.4 ± 0.1	60.5 ± 0.2
					Cristobalite	3.0L	3.0	94.0
31	SiO2	1250	1100	72	Mullite	1.1L	77.2	21.7
					Cristobalite + Tridymite (Si1-x(CuAl)xO2)			
					Liquid	71.2 ± 0.3	4.3 ± 0.0	24.5 ± 0.3
					Cristobalite	0.60 ± 0.06L	0.70	98.7
35	SiO2	–	1200	4	Tridymite	0.34 ± 0.07L	0.43	99.23
					Liquid	71.2 ± 0.5	4.3 ± 0.1	24.4 ± 0.4
					Cristobalite	0.42 ± 0.06L	0.60	99.0
					Tridymite	0.34 ± 0.04L	0.41	99.25
20	SiO2	1220	1200X	4	Liquid	73.7 ± 0.6	3.1 ± 0.1	23.2 ± 0.4
					Cristobalite	0.48 ± 0.06L	0.50	99.02
					Tridymite	0.34 ± 0.04L	0.37	99.29
					Liquid	77.5 ± 0.26	2.1 ± 0.1	20.4 ± 0.2
24	SiO2	1220	1200X	4	Cristobalite	0.39 ± 0.04L	0.40	99.21
					Tridymite	0.32 ± 0.04L	0.33	99.35
					Cuprite (Cu2O) + Delafossite (CuAlO2)			
					Liquid	35.8 ± 0.3	19.9 ± 0.3	44.3 ± 0.6
29	SiO2	1300	1100	48	Cuprite	99.97	0.00	0.03
					Delafossite	49.6	50.3	0.09
					Delafossite (CuAlO2) liquidus			
					Liquid	38.4 ± 0.6	23.7 ± 0.5	37.9 ± 0.7
29	SiO2	1300	1200	48	Delafossite	50.1	49.8	0.12
					Liquid	46.1 ± 0.1	18.7 ± 0.1	35.2 ± 0.1
					Delafossite	49.7	50.1	0.17
					Delafossite (CuAlO2) + Mullite (Al6+2xSi2-2xO13-x)			
12	SiO2	1200	1150	16	Liquid	28.2 ± 0.6	23.5 ± 0.1	48.2 ± 0.5
					Delafossite	50.5	49.4	0.15
					Mullite	2.1F	76.6	21.3
					Mullite (Al6+2xSi2-2xO13-x) liquidus			
8	Al2O3	–	1650	1	Liquid	19.1 ± 0.3	39.8 ± 0.5	41.1 ± 0.4
					Mullite	0.89F	76.8	22.3
					Liquid	20.3 ± 0.7	32.7 ± 0.5	47.1 ± 1.2
					Mullite	0.3L	78.0	21.7
6	Al2O3	1520	1500	2	Liquid	21.4 ± 0.2	28.9 ± 0.2	49.7 ± 0.2
					Mullite	1.2L	77.6	21.3
					Liquid	24.3 ± 0.4	27.9 ± 0.6	47.9 ± 0.9
					Mullite	1.3F	76.6	22.0
37	SiO2	–	1300	4	Tridymite (Si1-x(CuAl)xO2) liquidus			
					Liquid	78.1 ± 0.6	1.4 ± 0.0	20.5 ± 0.6
					Tridymite	0.33L	0.30	99.37
					Liquid	78.6 ± 0.5	1.3 ± 0.0	20.2 ± 0.4
36	SiO2	–	1200	4	Tridymite	–	–	–
					Liquid	76.4 ± 0.6	2.9 ± 0.1	20.7 ± 0.5
					Tridymite	0.37L	0.37	99.26

Note: ^F indicates that these values are believed to be affected by the secondary fluorescence; ^L indicates that Cu L_α-line was used for measurement, with subsequent correction according to equations (1)–(4). For all other cases, Cu K_α-line was used; ^x indicates samples studied with XRD.

slag. The samples were preheated 10 °C above the final temperature for 10 min to promote the formation of the initial liquid and then equilibrated for 4 h. The samples were then quenched, washed, and dried as described earlier. A piece of each sample was then taken for the following EPMA measurements and the rest of it was crushed into a powder using agate mortar and pestle and studied with X-ray diffraction (XRD).

The phase assemblage of selected samples was confirmed using Bruker D8 Advance X-ray diffractometer (Bruker Corporation, Billerica, Massachusetts, USA) with a LynxEye detector operating with the Cu K radiation ($\lambda = 1.5406 \text{ \AA}$) at 40 kV and 40 mA. Data were collected in a range from 15° to 70° with a speed of $0.5^\circ \cdot \text{min}^{-1}$. The phases present were identified from the XRD peaks using DIFFRAC.EVA V5.1 software and PDF-4+ 2021 database by the International Centre for Diffraction Data.

To ensure the achievement of equilibrium in the samples, the “4-point test” [34,55] was used including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of the phases by EPMA; (3) approaching the final equilibrium point from different starting conditions; and, importantly, (4) consideration of reactions specific to this system that may affect the achievement of equilibrium or reduce the accuracy.

It was found that equilibration for 2 h was sufficient for the achievement of equilibrium at the temperatures above 1200 °C. For the lower temperatures the equilibration time required was found to be significantly longer (Table 1) increasing up to 168 h for the sample equilibrated at 1060 °C. This was set as the lower temperature limit of the study since the decrease in temperature would lead to the irrational increase in the required equilibration time. Homogeneity of the phases was examined in different locations of the samples and can be verified by the standard deviation of the measurements (as given in Table 1). Different starting composition and substrates were used at several different temperatures (Table 1) to confirm the achievement of equilibrium; the compositions of liquids measured agreed with each other within the standard deviation of measurements. The only reaction that could possibly affect the phase equilibria in the system was the interaction between the sample and the substrate leading to the dissolution of the substrate material in the liquid phase; the compositional trends corresponding to the progress of this reaction were not observed for the samples reported in Table 1.

3. Thermodynamic optimization

The goal of the thermodynamic optimization is to obtain one set of model equations for the Gibbs free energies of all phases as functions of temperature, composition, and pressure through the evaluation of all available data on the system. Such model equations then allow all thermodynamic properties and phase equilibria of the system to be backcalculated. Thus, the data are rendered self-consistent and in agreement with thermodynamic principles.

FactSage 7.2 thermodynamic package [35] was used for modelling in the present study. The slag phase was described with the Modified Quasichemical formalism [56,57] that combines Gibbs energy parameters of two various models: 1) Quasichemical, as Modified Quasichemical Model (MQM), assuming formation of A-A, B-B, A-B pairs with Gibbs free energies expressed as a polynomial function of the “coordination-equivalent” fractions and 2) Bragg-Williams polynomial in terms of pure components concentrations for random mixing configuration entropy (B-W). The Modified Quasichemical Model takes into account second-nearest-neighbour short-range ordering of cations by considering pair exchange reactions;

$$(A-A) + (B-B) = 2(A-B); \Delta g_{AB} \quad (5)$$

The Gibbs free energy change for the quasichemical reaction 5 is used in the mathematical formulation of the model. The Gibbs energy of the solution is given by the following formula:

$$G = (n_A g_A^0 + n_B g_B^0) - T \Delta S^{\text{config}} + \left(\frac{n_{AB}}{2} \right) \Delta g_{AB}, \quad (6)$$

where g_A^0 and g_B^0 are the molar Gibbs free energies of the pure components, and ΔS^{config} is the configurational entropy of mixing given by randomly distributing the (A-A), (B-B) and (A-B) pairs:

$$\Delta S^{\text{config}} = -R(n_A \ln X_A + n_B \ln X_B) - R \left[n_{AA} \ln \left(\frac{X_{AA}}{Y_{AA}^2} \right) + n_{BB} \ln \left(\frac{X_{BB}}{Y_{BB}^2} \right) + n_{AB} \ln \left(\frac{X_{AB}}{2Y_A Y_B} \right) \right] \quad (7)$$

where X_A , X_B are overall mole fractions $X_A = 1 - X_B = n_A / (n_A + n_B)$, X_{AA} , X_{BB} and X_{AB} are pair fractions $X_{ij} = n_{ij} / (n_{AA} + n_{BB} + n_{AB})$, Y_A and Y_B are “coordination-equivalent” fractions: $Y_A = 1 - Y_B = \frac{Z_A n_A}{Z_A n_A + Z_B n_B}$

Finally, the excess Gibbs free energy of pair formation Δg_{AB} can be expanded as an empirical polynomial:

$$\Delta g_{AB} = \Delta g_{AB}^0 + \sum_{(i+j)>1} q_{AB}^{ij} Y_A^i Y_B^j \quad (8)$$

where Δg_{AB}^0 and q_{AB}^{ij} are empirical binary coefficients that are functions of temperature.

For the Bragg-Williams model Equation (6) takes the following shape:

$$G = (n_A g_A^0 + n_B g_B^0) + RT(n_A \ln X_A + n_B \ln X_B) + \frac{(n_A Z_A + n_B Z_B)}{2} Y_A Y_B \Delta g_{AB} \quad (9)$$

The excess Gibbs free energy of pair formation Δg_{AB} can then be expanded similar to Equation (8).

The binary parameters are extrapolated into a ternary system using geometric formalism developed by Pelton et al. [58,59] that combine Kohler (symmetrical) and Toop (asymmetrical) approaches depending on the chemistry of the components. Ternary model parameters are then introduced to reproduce the available phase equilibria data. The parameters $q_{AB(C)}^{ijk}$ describe the effect of the component C on the binary interaction between components A and B:

$$\begin{aligned} \text{Kohler (symmetrical)} : \Delta g_{AB} = & \left(\Delta g_{AB}^0 + \sum_{(i+j) \geq 1} q_{AB}^{ij} \left(\frac{Y_A}{Y_A + Y_B} \right)^i \left(\frac{Y_B}{Y_A + Y_B} \right)^j \right) \\ & + \sum_{(i+j+k) \geq 1} q_{AB(C)}^{ijk} \left(\frac{Y_A}{Y_A + Y_B} \right)^i \left(\frac{Y_B}{Y_A + Y_B} \right)^j Y_C^k \end{aligned} \quad (10)$$

$$\begin{aligned} \text{Toop (asymmetrical)} \Delta : g_{AB} = & \left(\Delta g_{AB}^0 + \sum_{(i+j) \geq 1} q_{AB}^{ij} \left(\frac{Y_A}{Y_A + Y_B} \right)^i \left(\frac{Y_B}{Y_A + Y_B} \right)^j \right) \\ & + \sum_{(i+j+k) \geq 1} q_{AB(C)}^{ijk} Y_A^i (1 - Y_A)^j \left(\frac{Y_C}{Y_B + Y_C} \right)^k \end{aligned} \quad (11)$$

In the course of the present study only Toop (asymmetrical) approach was used.

4. Results and discussion

4.1. Experimental results

Examples of typical microstructures observed in the samples in the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper quenched from the equilibration temperatures are given in Fig. 1. All micrographs were obtained using backscattered electron microscopy (BSE) to provide contrast based on mean atomic number. All

Table 2Optimized thermodynamic parameters of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system.

a) Stoichiometric compounds					
Compound	ΔH_{298}^0 , J/mol	S_{298}^0 , J/(mol·K)	Temperature range, K	$C_p(T)$, J/(mol·K)	Ref.
Tridymite (high), SiO ₂	−907045	45.524	298–1991	$75.373 - 5958095T^{-2} + 958246122T^{-3}$	[62]
Cristobalite (high), SiO ₂	−906377	46.029	298–1996	$83.514 - 2455360T^{-2} + 280072194T^{-3} - 374.693T^{-0.5}$	[62]
Corundum, Al ₂ O ₃	−1675700	50.82	0–298	$-8.855266 \cdot 10^{-5}T^2 + 2.394632 \cdot 10^{-5}T^3$	[30]
			298–2327	$-1.179747 \cdot 10^{-7}T^4 + 1.631416 \cdot 10^{-10}T^5$	
				$155.019 - 3861363T^{-2} + 409083646T^{-3}$	[62]
				$-828.387T^{-0.5}$	
Cuprite, Cu ₂ O	−170258	92.68	298–3000	$66.26 + 0.01592T - 748000T^{-2}$	[8]
Delafossite, CuAlO ₂	−936141	64.41	298–1600	$85.8445 + 0.02353T - 1974660T^{-2} - 4.09025 \cdot 10^{-6}T^2$	Present study
b) Solid solutions					
Mullite, Al ₂ ³⁺ (Al ³⁺ , Si ⁴⁺)(O ^{2−} , Va) ₅					
$G_{Al_2AlO_5}^{-1} = 1.5G_{Al_2O_3} + \Delta_{Al_2Al}$;					[64]
$G_{Al_2O_3} = G_{Al_2O_3}^{(corundum)} + 0.57$;					[64]
$\Delta_{Al_2Al} = 86508.4 - 0.4184T$;					[64]
$G_{Al_2AlVa}^{+9} = 1.5G_{Al_2O_3} + \Delta_{Al_2Al}$;					[64]
$G_{Al_2SiO_5} = G_{Al_2SiO_5}^0 + 12300.96 - 1.4644T$					[30]
$G_{Al_2SiVa}^{+10} = G_{Al_2SiO_5} + G_{Al_2Si}$;					[64]
$\Delta_{Al_2Si} = 0$;					[64]
${}^0L_{AlAlSi:O} = {}^0L_{AlAlSi:Va} = -83680$					[30]
Endmember	ΔH_{298}^0 , J/mol	S_{298}^0 , J/(mol·K)	Temp. range, K	$C_p(T)$, J/(mol·K)	Ref.
Al ₂ SiO ₅	−2587770	95.79	298–1473 1473–3000	$188.22 + 0.014677T - 1.236 \cdot 10^7T^{-2} + 1.6024 \cdot 10^{10}T^{-3} + 3205.2T^{-1}$ $188.22 + 0.014677T - 1.236 \cdot 10^7T^{-2} + 1.6024 \cdot 10^{10}T^{-3} + 16549T^{-1}$	[64]
c) Liquid slag					
Liquid slag effective coordination numbers:					
$Z_{Cu^{II+}}^{Cu^{II+}} = Z_{Cu^{II+}}^{Al^{III+}} = Z_{Cu^{II+}}^{Si^{IV+}} = 0.6887$,					
$Z_{Al^{III+}}^{Al^{III+}} = Z_{Al^{III+}}^{Cu^{II+}} = Z_{Al^{III+}}^{Si^{IV+}} = 2.0662$,					
$Z_{Si^{IV+}}^{Si^{IV+}} = Z_{Si^{IV+}}^{Cu^{II+}} = Z_{Si^{IV+}}^{Al^{III+}} = 2.7549$,					
Endmember	ΔH_{298}^0 , J/mol	S_{298}^0 , J/(mol·K)	Temp. range, K	$C_p(T)$, J/(mol·K)	Ref.
SiO ₂	−896796	50.83	298–1996	$83.514 - 2455360T^{-2} + 280072194T^{-3}$ $-374.693T^{-0.5}$	[62]
Cu ₂ O	−120900	120.958	298–900 900–1350 1350–3000	$71.0551 + 0.014562364T - 749586T^{-2}$ $8.576166 + 0.081738T$ 120	[31]
Al ₂ O ₃	−1594000	50.851	0–298 298–350 350–593 593–2327	$-8.855266 \cdot 10^{-5}T^2 + 2.394632 \cdot 10^{-5}T^3 - 1.179747 \cdot 10^{-7}T^4 + 1.631416 \cdot 10^{-10}T^5$ $-298.66 + 1.266T$ 144.6321 $179.36549 - 0.009192T + 975341T^{-2} - 828.387T^{-0.5} + 409083646T^{-3}$	[30] [62]
Excess parameters:					
Excess interaction parameter	Value, J/mol	Model	Ref.		
$\Delta g_{Al^{III+}, Si^{IV+}}^0$	5732.08	Quasichemical	[30]		
$q_{Al^{III+}, Si^{IV+}}^{10}$	4184	Quasichemical			
$q_{Al^{III+}, Si^{IV+}}^{03}$	101252.8	Quasichemical			
$q_{Al^{III+}, Si^{IV+}}^{05}$	−128867.2	Quasichemical			
$q_{Al^{III+}, Si^{IV+}}^{07}$	63596.8	Quasichemical			
$\Delta g_{Al^{III+}, Cu^{II+}}^0$	54810.4	Bragg-Williams	Present study		
$q_{Al^{III+}, Cu^{II+}}^{10}$	21903.24	Bragg-Williams			
$\Delta g_{Si^{IV+}, Cu^{II+}}^0$	40333.76	Bragg-Williams	[63]		
$q_{Si^{IV+}, Cu^{II+}}^{50}$	237107.28–99.1608T	Bragg-Williams			
$q_{Si^{IV+}, Cu^{II+}}^{70}$	27279.68	Bragg-Williams			
$q_{Si^{IV+}, Cu^{II+}}^{01}$	−47362.88 + 20.92T	Bragg-Williams			
$q_{Al^{III+}, Si^{IV+}, Cu^{II+}}^{007}$	−16736	Bragg-Williams	Present study		
$q_{Si^{IV+}, Cu^{II+}, Al^{III+}}^{001}$	−10250.8	Quasichemical			
$q_{Si^{IV+}, Cu^{II+}, Al^{III+}}^{031}$	−66358.24	Quasichemical			
$q_{Al^{III+}, Si^{IV+}, Cu^{II+}}^{105}$	−50208	Bragg-Williams			
$q_{Si^{IV+}, Cu^{II+}, Al^{III+}}^{201}$	−164180.16	Quasichemical			
$q_{Si^{IV+}, Cu^{II+}, Al^{III+}}^{022}$	−99997.6	Quasichemical			

micrographs demonstrated the coexistence of liquid copper metal with liquid and solid oxide phases of the system. The phases illustrated in these figures include corundum (Al₂O₃) (Fig. 1a and b), cristobalite (Si_{1-x}(CuAl)_xO₂) (Fig. 1c–g), cuprite (Cu₂O) (Fig. 1e,j), delafossite (CuAlO₂) (Fig. 1i–k), mullite (Al_{6+2x}Si_{2-2x}O_{13-x}) (Fig. 1f,k,l), tridymite (Si₁₋

_x(CuAl)_xO₂) (Fig. 1g and h) and the coexistence of two liquids (Fig. 1b, d).

The average compositions of the phases obtained in the present study were reported in Table 1. The solubilities of Si and Al in the metallic copper were found to be negligible (<0.05 mol.%, 0 mol.% in most of

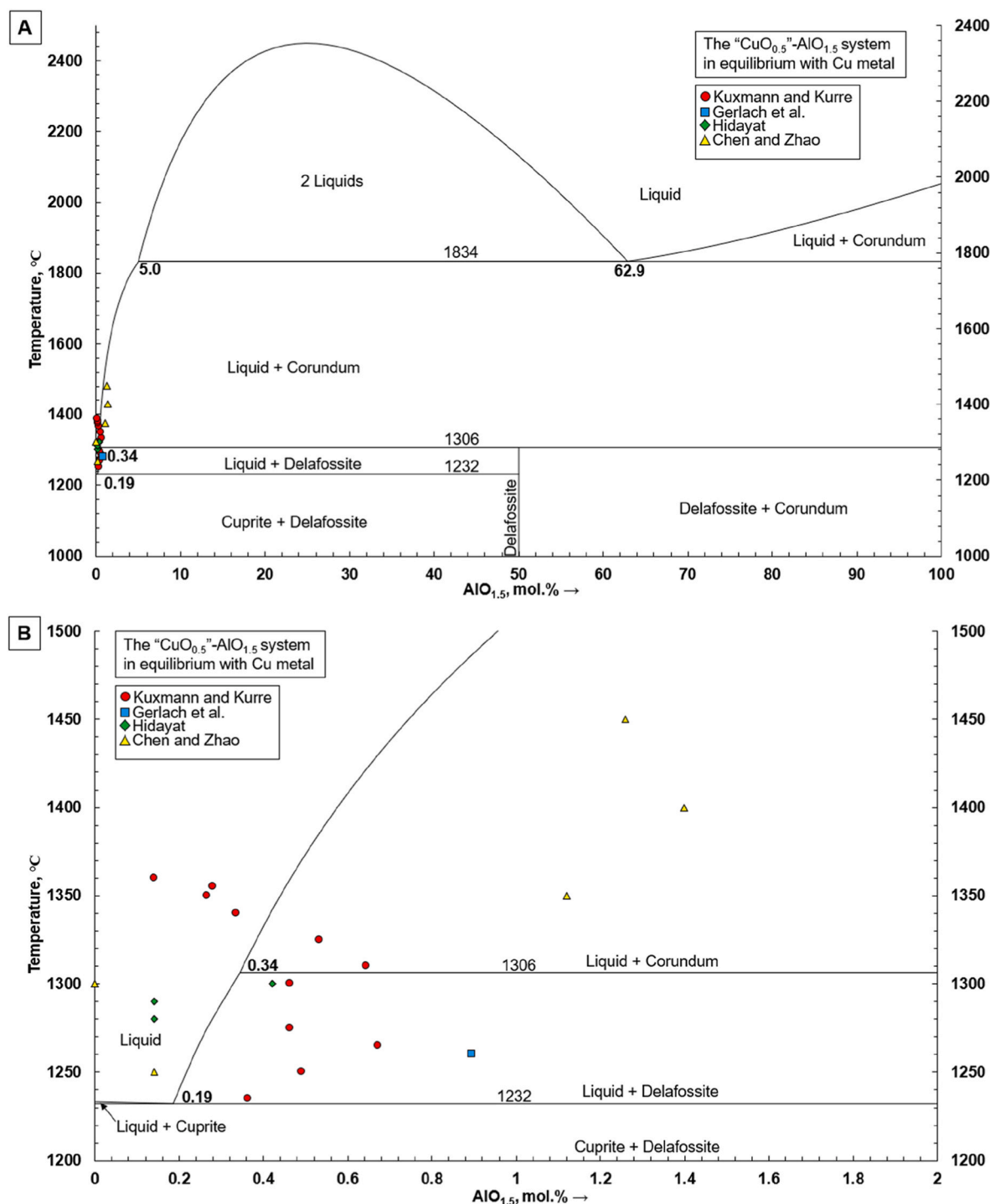


Fig. 2. Modelled phase diagram of the "CuO_{0.5}"-AlO_{1.5} system in equilibrium with metallic copper with experimental data from Refs. [12,18,19,65]: (a) full diagram; and (b) detailed "CuO_{0.5}"-rich area of the diagram.

the cases) for every sample and were not included in Table 1. The results of this study and the experimental data reported by previous authors [14,19], were used for the thermodynamic optimization of the "CuO_{0.5}"-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper.

4.2. Thermodynamic optimization

No information on the previous optimization attempts of the "CuO_{0.5}"-AlO_{1.5} or "CuO_{0.5}"-AlO_{1.5}-SiO₂ system was found in the literature; the presented study then provides the first optimization of the

indicated systems. The optimized thermodynamic parameters of the "CuO_{0.5}"-AlO_{1.5}-SiO₂ system are listed in Table 2. Three-sublattice compound energy model of mullite Al₂³⁺(Al³⁺,Si⁴⁺)(O²⁻,Va)₅ was adopted from Hillert and Jonsson [60] and Dumitrescu and Sundman [61] with the parameters accepted from Shevchenko and Jak [30]. Corundum (Al₂O₃) [30,62], cuprite (Cu₂O) [8,31] and delafossite (CuAlO₂) were treated as stoichiometric compounds. Tridymite and cristobalite (SiO₂) [62] were treated as stoichiometric compounds, but shown by the present experimental study to form a solution in the presence of the AlO_{1.5} and "CuO_{0.5}" impurities up to 3 mol.% AlO_{1.5} and

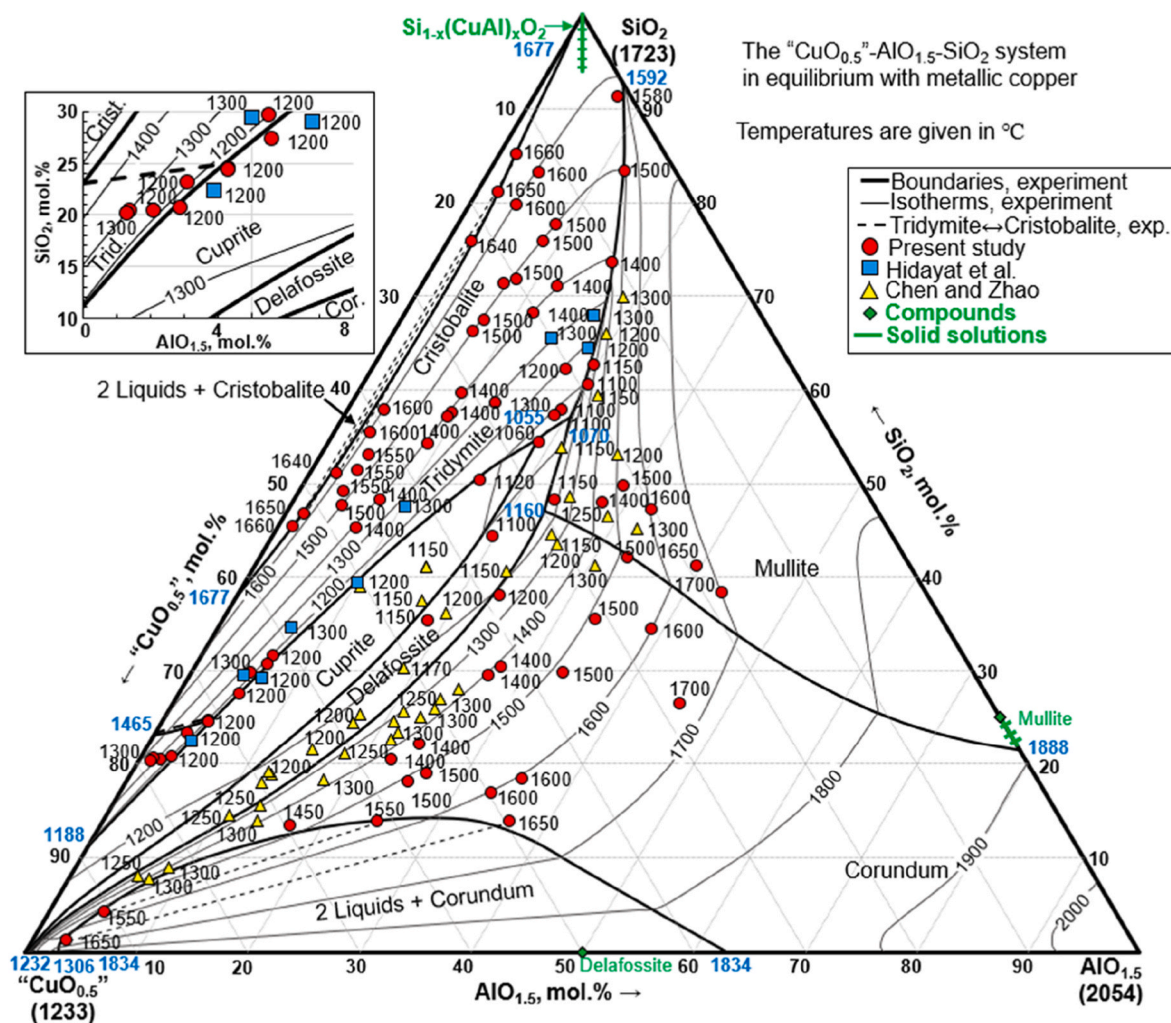


Fig. 3. The experimental liquidus surface of the "CuO_{0.5}"-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper. Experimental data by Hidayat et al. [14] and Chen and Zhao [19]. Temperatures are given in °C.

3 mol.% "CuO_{0.5}", respectively. The binary "CuO_{0.5}"-SiO₂ and AlO_{1.5}-SiO₂ interaction parameters were adopted from the previous studies [30, 63]. The binary "CuO_{0.5}"-AlO_{1.5} and ternary "CuO_{0.5}"-AlO_{1.5}-SiO₂ slag interaction parameters and those for stoichiometric delafossite (CuAlO₂) were optimized in the present study. Phase diagram of the "CuO_{0.5}"-AlO_{1.5} system in equilibrium with metallic copper is given and compared with the results of previous studies [12,18,19] in Fig. 2. Experimental liquidus projection of the "CuO_{0.5}"-AlO_{1.5}-SiO₂ is given and compared with the experimental results of this and previous studies [14,19] in Fig. 3. Experimental boundaries and isotherms included in Fig. 3 are mainly based on the results of this study (Table 1); for those regions where no experimental results were obtained by the authors, the data from previous studies [14,19] were used.

4.3. The "CuO_{0.5}"-AlO_{1.5} system in equilibrium with metallic copper

The literature data on the phase equilibria in this system have discrepancies: the liquid compositions presented by Chen and Zhao [19] include liquid phase reported to contain 100 mol.% "CuO_{0.5}" while being in equilibrium with corundum (Al₂O₃). The liquid compositions reported by Kuxmann and Kurre [12] are not self-consistent. It can be seen from Fig. 2 that there is a significant discrepancy in the existing experimental data on the phase equilibria within the "CuO_{0.5}"-AlO_{1.5}; one of the reasons for this disagreement is the bad quenching of the "CuO_{0.5}"-rich liquids due to their low viscosity. At the same time the

qualitative change in the solid phase present in equilibrium with liquid at different temperatures allows to determine the temperature of invariant reactions. There was no information on the compositions of liquid phases taking part in the invariant reactions of the system given in previous studies [12,18,19,65].

The decomposition of delafossite (CuAlO₂) into corundum (Al₂O₃) and liquid phase was reported by Hidayat [65] to proceed in the temperature range from 1290 to 1300 °C. The following correlation between the Gibbs free energy of CuAlO₂ formation from the corresponding pure solid oxides (cuprite (Cu₂O) and corundum (Al₂O₃)) and temperature, based on the experimental data reported in previous studies [66–71], was used in the present study:

$$\Delta_{f(\text{ox})} G_{\text{CuAlO}_2}^0 = -12.84 + 7.10 \times 10^{-3} T, \text{ kJmol}^{-1} \quad (12)$$

Fig. 4 provides the graphical comparison of the present correlation and those reported in previous studies [66–71]. The existing data is scattered and cannot be described by one correlation.

The data on the presence of two immiscible liquids over corundum (Al₂O₃) in the SiO₂-poor area of the system indicated that there should be the "CuO_{0.5}"-poor liquid ↔ Corundum (Al₂O₃) + "CuO_{0.5}"-rich liquid monotectic reaction in the "CuO_{0.5}"-AlO_{1.5} system in equilibrium with metallic copper. The temperature of this reaction and the compositions of liquid phases taking part in it could not be measured directly and were therefore predicted based on the existing data. The phase diagram of the "CuO_{0.5}"-AlO_{1.5} system in equilibrium with metallic copper presented in

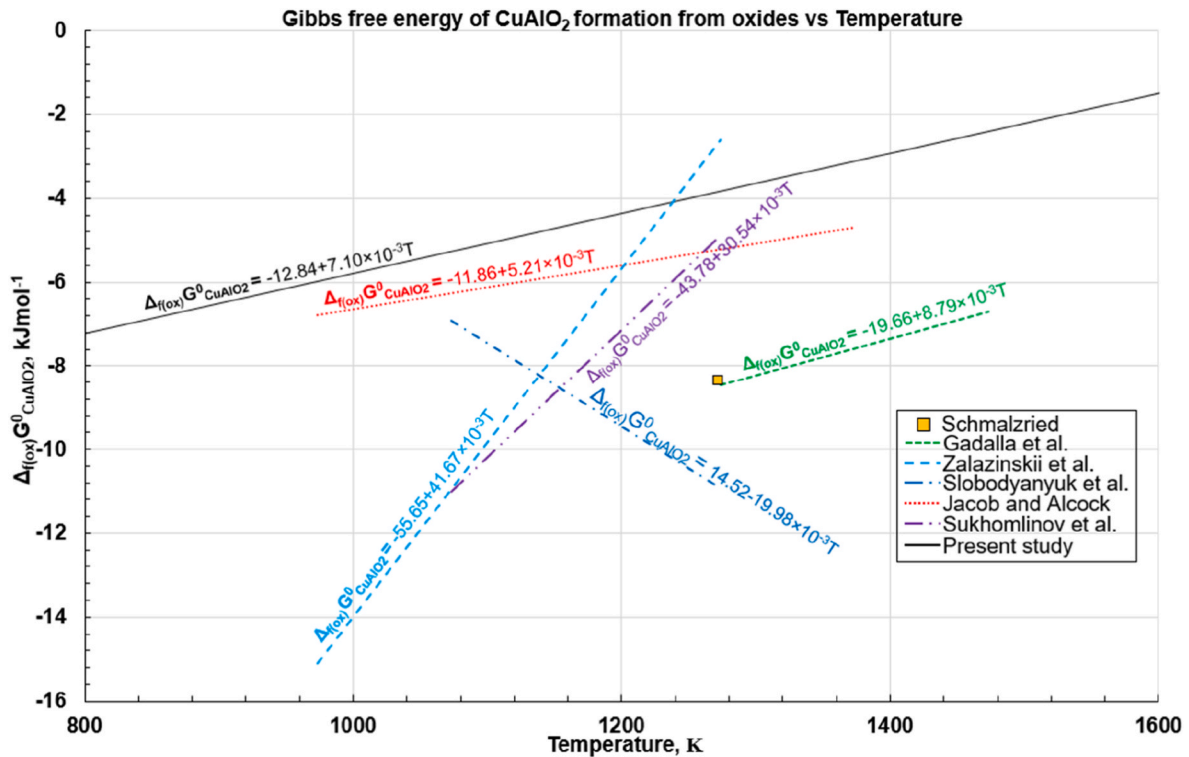


Fig. 4. The Gibbs free energy of delafossite (CuAlO_2) formation from the corresponding pure solid oxides (Cu_2O and Al_2O_3) compared with the results from previous studies [66–71].

Table 3

Invariant reactions of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ system in equilibrium with metallic copper.

Temperature, °C	Type	Reaction	Liquid composition, mol. %	
			“ $\text{CuO}_{0.5}$ ”	$\text{AlO}_{1.5}$
1232	Eutectic	Liquid $\leftrightarrow$ Cuprite (Cu_2O) + Delafossite (CuAlO_2)	99.81	0.19
1306	Peritectic	Liquid + Corundum (Al_2O_3) $\leftrightarrow$ Delafossite (CuAlO_2)	99.66	0.34
1834	Monotectic	“ $\text{CuO}_{0.5}$ ”-poor liquid $\leftrightarrow$ Corundum (Al_2O_3) + “ $\text{CuO}_{0.5}$ ”-rich liquid	95.0 36.1	5.0 62.9

Fig. 2 was then based on estimated data and should be used with caution as it might not present accurate information on phase equilibria of the system.

Three invariant reactions were predicted in the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ system in equilibrium with metallic copper; their temperatures and compositions of liquid phase taking part in them in Table 3. The

predicted eutectic and peritectic reactions are similar to those shown by Chen and Zhao [19]. The existence of the monotectic reaction predicted at 1834 °C was confirmed by the experimental results in the low- SiO_2 area of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 system in equilibrium with metallic copper (Table 1).

4.4. Liquidus surface of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 system in equilibrium with metallic copper

The thermodynamic parameters were optimized so to minimise the discrepancy between modelling predictions and the experimental data available for the liquidus of the system. The liquidus of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 system in equilibrium with metallic copper is described by the primary phase fields of corundum (Al_2O_3), cristobalite and tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$), cuprite (Cu_2O), delafossite (CuAlO_2) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) and two 2-liquid miscibility gaps located in cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and corundum (Al_2O_3) primary phase fields. The area of the system with the concentrations of $\text{AlO}_{1.5} > 20$ mol.% is characterized by the primary phase fields of corundum (Al_2O_3) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) only.

The predicted liquidus of cristobalite/tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and

Table 4

Comparison of the invariant reactions of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 system in equilibrium with metallic copper.

Temperature, °C	Type	Reaction	Liquid composition, mol.%			Ref.
			“ $\text{CuO}_{0.5}$ ”	$\text{AlO}_{1.5}$	SiO_2	
1055	Eutectic	Liquid $\leftrightarrow$ Cuprite (Cu_2O) + Delafossite (CuAlO_2) + Tridymite (SiO_2)	27.0	19.0	54.0	Present study, exp.
1055			33.0	17.3	49.8	Present study, model
1090			23.0	20.3	56.7	[19]
1070	Quasi-peritectic	Liquid + Mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) $\leftrightarrow$ Delafossite (CuAlO_2) + Tridymite (SiO_2)	22.5	20.5	57.0	Present study, exp.
1067			28.4	18.6	53.0	Present study, model
1160	Quasi-peritectic	Liquid + Corundum (Al_2O_3) $\leftrightarrow$ Delafossite (CuAlO_2) + Mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$)	30.0	23.0	47.0	Present study, exp.
1177			37.1	20.1	42.8	Present study, model
1110	Quasi-peritectic	Liquid + Corundum (Al_2O_3) $\leftrightarrow$ Delafossite (CuAlO_2) + Tridymite (SiO_2)	16.3	22.3	61.3	[19]
1130	Quasi-peritectic	Liquid + Mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) $\leftrightarrow$ Corundum (Al_2O_3) + Tridymite (SiO_2)	18.1	22.4	59.5	[19]

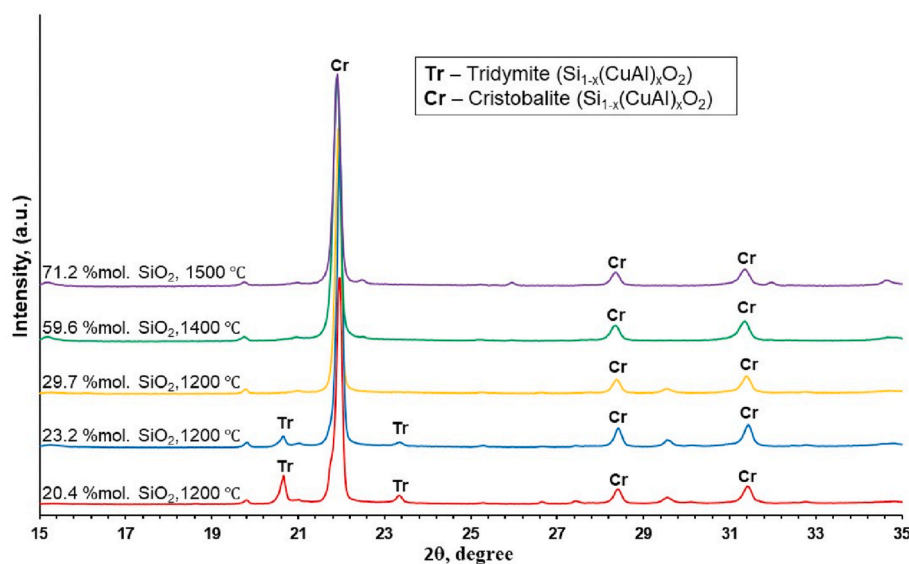


Fig. 5. XRD patterns of the samples containing tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) (a and b) and/or cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) (a–e) equilibrated at different temperatures.

cuprite (Cu_2O) and boundary lines between cristobalite/tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and cuprite (Cu_2O), cristobalite/tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) and, partially, corundum (Al_2O_3) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) are in a good agreement with the experimental results of the present study and those reported by Hidayat et al. [14]. The most significant difference between the model and the experiment can be observed in the middle part of the liquidus projection where all 3 invariant reactions of the system take place. This difference varies from 10 to 190 K depending on the area. The difference between the model and the experiment for this region could not be decreased without affecting the accuracy of the predictions in other areas of the diagram. The final set of the parameters then provides an optimal agreement of the model with the experimental data in the $\text{AlO}_{1.5}$ -poor and the $\text{AlO}_{1.5}$ -rich areas of the system. To increase the accuracy of the model, the parameters of the binary sub-systems, and potent should be reoptimized in the future study.

Three invariant reactions predicted to proceed in the system were described and compared with experimental data of the present study and those proposed by Chen and Zhao [19] in Table 4. The calculated temperatures of the invariant reactions are in a good agreement with the experimental findings. At the same time, the predicted compositions of the liquid phases taking part in the proposed invariant reactions are to some degree different to the experimental data. Similarly to the whole calculated liquidus surface of the system, the compositions of the eutectic and peritectic liquids were an optimal agreement of the model with the experimental data in the $\text{AlO}_{1.5}$ -poor and the $\text{AlO}_{1.5}$ -rich areas of the system; the difference with experimental data indicate the necessity of the further investigation of binary sub-systems. The difference with the findings by Chen and Zhao [19] is explained by the insufficiency of the experimental data in the discussed region obtained by the authors.

The delafossite (CuAlO_2) primary phase field is characterized by a narrow interval of stability. No data on the liquidus temperatures in the primary phase field of cuprite (Cu_2O) were obtained except those on the cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) + cuprite (Cu_2O) and cuprite (Cu_2O) + delafossite (CuAlO_2) boundaries; for the liquidus of cuprite (Cu_2O) the experimental results reported by Chen and Zhao [19] were accepted.

The immiscibility gap in the low- SiO_2 area of the diagram is characterized by a wide temperature range of existence; the minimum point on the gap was predicted to be at ~ 78 mol.% “ $\text{CuO}_{0.5}$ ”, 10 mol.% $\text{AlO}_{1.5}$ and 12 mol.% SiO_2 at a temperature slightly lower than 1400 °C. The experimental results showing 2 liquids in equilibrium with corundum (Al_2O_3) (Table 1) confirm the existence of this immiscibility gap in both

discussed ternary and “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ binary system, however indicate that the minimum point on the gap should be between 1450 and 1550 °C. The immiscibility gap in the primary phase field of cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) is predicted to have a temperature minimum between 1500 and 1550 °C. The experimental results show that the interval of stability of this gap is more narrow than predicted, having temperature minimum between 1600 and 1640 °C.

The liquidus in the cristobalite and tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) primary phase fields was measured in the present study at temperatures ranging from 1060 to 1660 °C. The tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) was found to be stable only at the concentrations of “ $\text{CuO}_{0.5}$ ” > ~ 70 mol.%; The experimental data obtained for the liquidus isotherms at 1600, 1500, 1400 and 1100 °C were consistent with the topology of those reported previously at 1300 and 1200 °C by Hidayat et al. [14] and the thermodynamic predictions. Points from Hidayat et al. [14] may be re-identified to contain cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) rather than tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) based on the present results.

Fig. 8 provides the predicted liquidus surface of the “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ – SiO_2 system in equilibrium with metallic copper compared with the experimental finding of the present study and previous authors (Hidayat et al. [14] and Chen and Zhao [19]).

4.5. Tridymite-cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) transition

The identities of the tridymite and cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) phases were distinguished on the basis of their different crystal morphologies. Fig. 1c–g shows the rounded liquid/crystal interface in the case of the cubic high-temperature form of cristobalite in contrast to the faceted plate-shaped tridymite (SiO_2) crystals formed at low temperature (see Fig. 1g and h). The correspondence between the observed shape of the crystal and the phase (tridymite or cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$)) was confirmed after 5 samples belonging to different areas of the diagram were quenched from different temperatures (1200, 1400 and 1500 °C) and studied using X-ray powder diffraction. The compositions of phases present in these samples are given in Table 1. Fig. 5 shows the comparison of the XRD patterns of the samples. The samples with only cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) present initially had a structure similar to that shown in Fig. 1c, and those with both tridymite and cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) similar to Fig. 1g. The volume fraction of the tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) observed in the samples correlated with the intensity of the diffraction peaks identified as tridymite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$).

Unlike other systems with SiO_2 [72–74] the tridymite–cristobalite

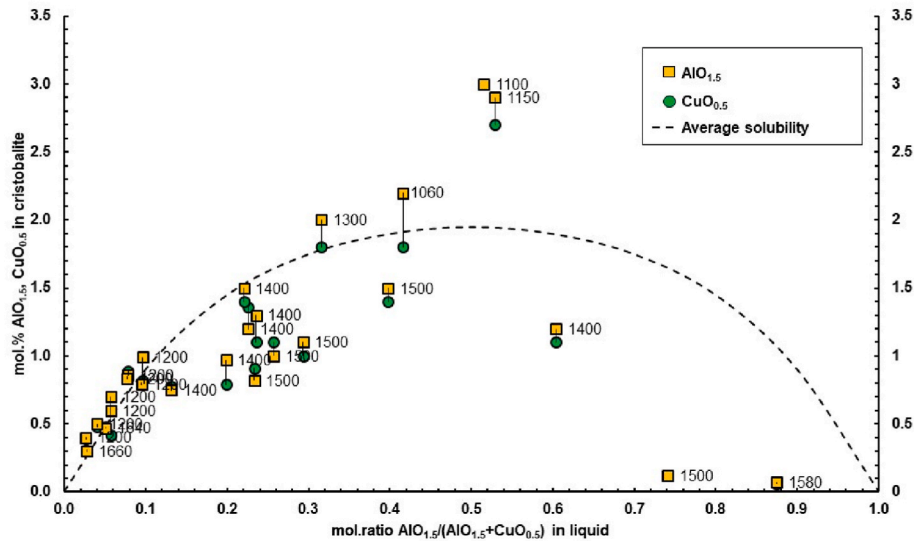


Fig. 6. The correlation between the solubility of $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” in cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and the $\text{AlO}_{1.5}/(\text{AlO}_{1.5} + \text{“CuO}_{0.5}\text{”})$ ratio in liquid slag.

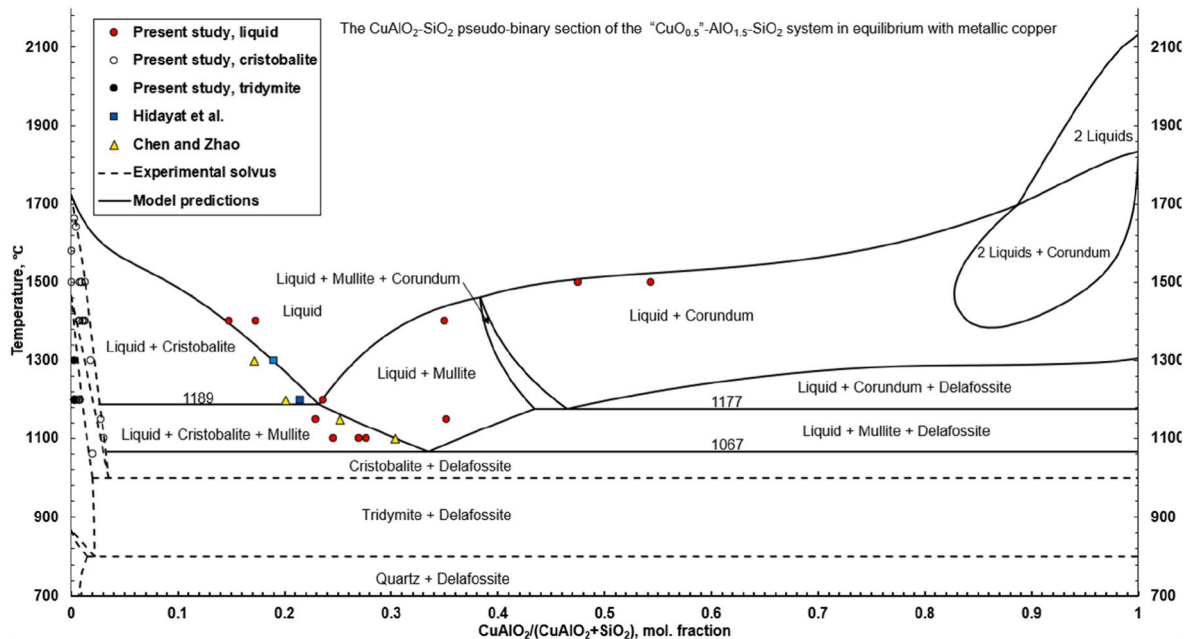


Fig. 7. The $\text{CuAlO}_2\text{-SiO}_2$ pseudo-binary section of the “ $\text{CuO}_{0.5}$ ”- $\text{AlO}_{1.5}$ - SiO_2 system in equilibrium with metallic copper. Solid lines are calculated with FactSage, dashed lines represent estimation based on the experimental data for tridymite and cristobalite.

($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) transition temperature was proved to vary with the composition of the liquid slag. This feature of the system is believed to appear due to the charge compensation effect [75], which also leads to significant solubilities of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ in tridymite and cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$).

The $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” concentrations in the tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$), after further custom corrections (equations (1)–(4)), were found to depend on the liquid composition increasing from negligible for the $\text{AlO}_{1.5}\text{-SiO}_2$ and “ $\text{CuO}_{0.5}$ ”- SiO_2 binary systems and reaching 3.0 and 3.0 mol.% $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” respectively for the concentration of 20.4 mol.% $\text{AlO}_{1.5}$ and 19.2 mol.% “ $\text{CuO}_{0.5}$ ” in the liquid slag. The average ratio of “ $\text{CuO}_{0.5}$ ”/(“ $\text{CuO}_{0.5}$ ” + $\text{AlO}_{1.5}$) in the tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) was found to equal 0.49 ± 0.03 . The compositions of cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) measured using Cu L_{α} lines were plotted in Fig. 3. The correlation between the solubility of $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” in cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and the $\text{AlO}_{1.5}/$

($\text{AlO}_{1.5} + \text{“CuO}_{0.5}\text{”}$) ratio in liquid slag is graphically presented in Fig. 6. The solubility of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ was shown to increase with the $\text{AlO}_{1.5}/(\text{AlO}_{1.5} + \text{“CuO}_{0.5}\text{”})$ ratio in liquid slag approaching 0.5; for the same $\text{AlO}_{1.5}/(\text{AlO}_{1.5} + \text{“CuO}_{0.5}\text{”})$ ratios the solubility increases with the decrease in the temperature. The possible error of the “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ solubility measurements was estimated to be below 20% based on the standard deviations of the obtained results. The data on the concentrations of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ in tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) are given to demonstrate the formation of the SiO_2 -based solid solution and preliminary describe the possible correlations between the solubility of these components with the composition of liquid phase. More accurate description of this effect will require longer sample equilibration times to prove the achievement of the equilibrium solubility of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ in tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and was outside of the scope of this study.

The solubility of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ in the tridymite/cristobalite

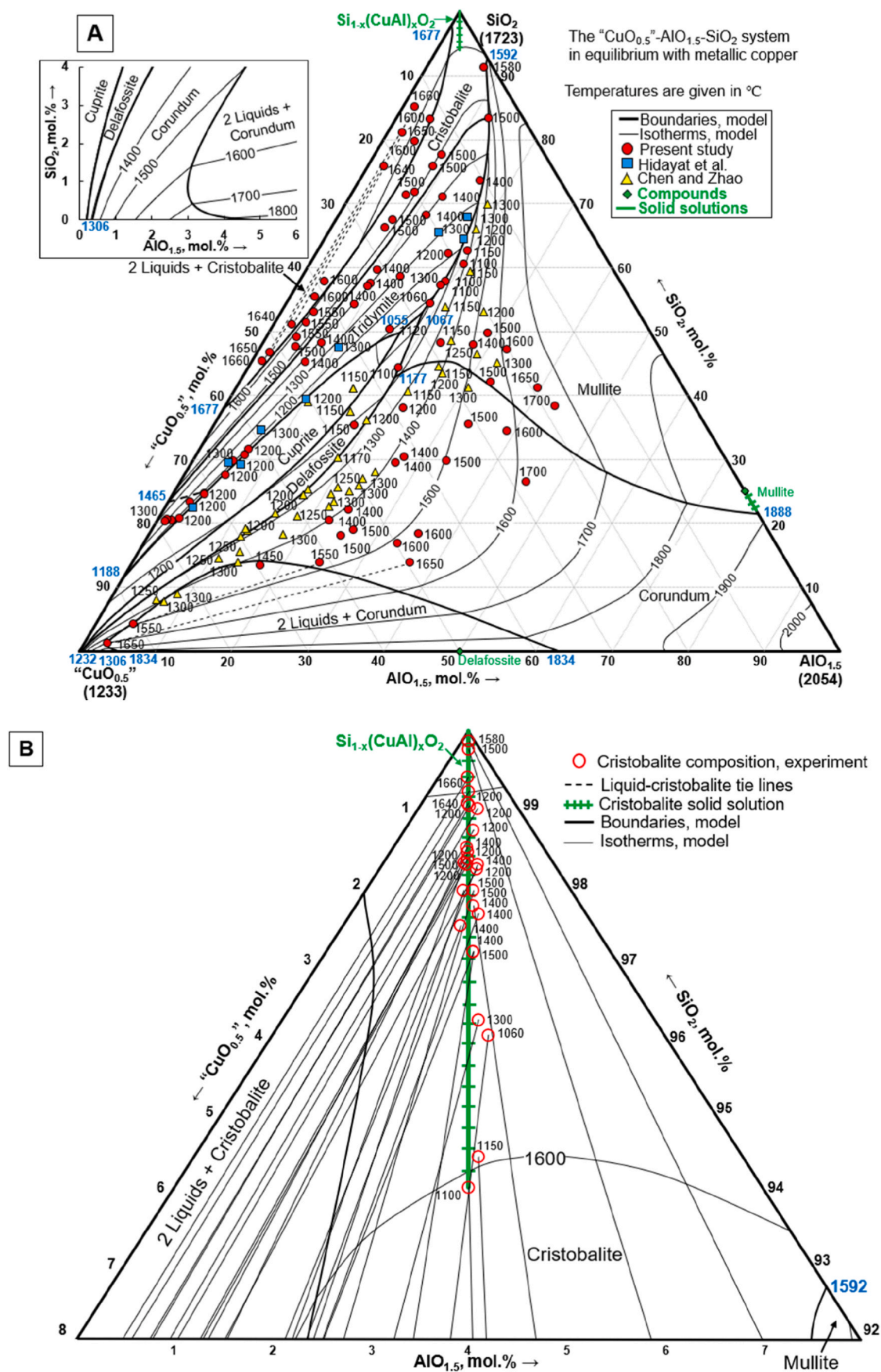


Fig. 8. The calculated liquidus surface of the “CuO_{0.5}”-AlO_{1.5}-SiO₂ system in equilibrium with metallic copper: (a) full modelled surface; and (b) detailed SiO₂-rich area with cristobalite (Si_{1-x}(CuAl)_xO₂) compositions. Experimental data by Hidayat et al. [14] and Chen and Zhao [19]. Temperatures are given in °C.

($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) was not modelled in the course of this study; this will be addressed in the future works. This effect was discovered during the presented study and was not one of its initial goals. The introduction of the parameters taking into account the solubility of “ $\text{CuO}_{0.5}$ ” and $\text{AlO}_{1.5}$ in the tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) will not affect the liquidus of the system except of the tridymite–cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) boundary.

The predicted CuAlO_2 – SiO_2 pseudo-binary section of the “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ – SiO_2 system in equilibrium with metallic copper is shown and compared with the existing experimental data in Fig. 7. The disagreement between the model and the experiment is explained by the experimental $\text{AlO}_{1.5}/(\text{AlO}_{1.5} + \text{“CuO}_{0.5}\text{”})$ ratios being different from 0.5 (constant for the described section). The cristobalite, tridymite and quartz ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) solid solutions are shown using dashed lines since there was no information on temperatures of their transitions obtained in the course of this study. Cristobalite is stabilised over tridymite by higher solubility of the “ CuAlO_2 ” endmember in it, so that the cristobalite–tridymite transition temperature is lowered from 1465 to <1050 °C. By analogy, some stabilisation of tridymite over quartz is predicted. The 2 liquids immiscibility presented in the high-temperature area of the CuAlO_2 –rich side of the pseudo-binary section progresses from the “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ binary system where it covers the “ $\text{CuO}_{0.5}$ ”–poor liquid $\leftrightarrow$ corundum (Al_2O_3) + “ $\text{CuO}_{0.5}$ ”-rich liquid monotectic reaction.

4.6. The solubilities of “ $\text{CuO}_{0.5}$ ” in corundum (Al_2O_3) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$)

The “ $\text{CuO}_{0.5}$ ” concentration in the corundum (Al_2O_3) in equilibrium with “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ – SiO_2 liquid measured with $\text{Cu K}\alpha$ lines was found to be overestimated compared to those measured with $\text{Cu L}\alpha$ lines (0.05 mol.%); the accurate measurement of “ $\text{CuO}_{0.5}$ ” concentration in the mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) in the present study was not possible due to the small size of the particles. The values of 0.69 ± 0.32 and 0.67 mol.% “ $\text{CuO}_{0.5}$ ” reported by Chen and Zhao [19] for corundum (Al_2O_3) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$), respectively, are believed to be not accurate and affected by the secondary X-ray fluorescence.

5. Conclusions

The study produced the new experimental data on phase equilibria in the “ $\text{CuO}_{0.5}$ ”– $\text{AlO}_{1.5}$ – SiO_2 system in equilibrium with metallic copper in the temperature range from 1060 to 1700 °C, including: the primary phase fields of tridymite and cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$), corundum (Al_2O_3), cuprite (Cu_2O), delafossite (CuAlO_2) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$), as well as the two immiscibility gaps in the low- and high-silica areas of the diagram.

The liquidus surface of the system, predicted based on the newly optimized model parameters, was shown to be dominated by the primary phase fields of corundum (Al_2O_3), cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) and mullite ($\text{Al}_{6+2x}\text{Si}_{2-2x}\text{O}_{13-x}$) and to contain 2 invariant quasi-peritectics and an invariant eutectic; the approximate temperatures and the compositions of liquids taking part in these invariants were reported.

The temperature of tridymite–cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) transition was shown to vary from 1465 to ~1170 °C depending on the composition of the liquid slag. The correlation between the concentration of $\text{AlO}_{1.5}$ in the liquid slag and the solubilities of $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” in the tridymite/cristobalite ($\text{Si}_{1-x}(\text{CuAl})_x\text{O}_2$) was reported; the maximum solubilities were measured equal to 3.0 and 3.0 mol.% $\text{AlO}_{1.5}$ and “ $\text{CuO}_{0.5}$ ” respectively for the concentration of 20.4 mol.% $\text{AlO}_{1.5}$ and 19.2 mol.% “ $\text{CuO}_{0.5}$ ” in the liquid slag.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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